

PAPER_692: M51 Whirlpool Galaxy: Tidal Arm Formation and UQFF Star Formation

Class: M51WhirlpoolTidalInteraction

CP4 Entry: #276

Keywords: M51, Whirlpool galaxy, tidal arms, spiral pitch angle, SFR, tidal force, UQFF

Session: 174 | **Version:** v5.31

Source: grok_share_ba508f76c8e.txt

Abstract

The Whirlpool Galaxy (M51a / NGC 5194) and its companion M51b (NGC 5195) form a spectacular interacting pair 23 Mly away. Tidal forces from M51b drive the prominent spiral arm structure and enhance star formation rates.

1. System Parameters

Parameter	Value
M_{M51a}	$1.6 \times 10^{11} M_{\odot}$
M_{M51b}	$2 \times 10^{10} M_{\odot}$
Separation	7.5 kpc
SFR	$3 M_{\odot}/\text{yr}$

2. Tidal Force

$$F_{tidal} = \frac{2GM_{comp}R_d}{d_{sep}^3}$$

This differential force stretches M51a's disk and drives arm formation.

3. Spiral Arm Pitch Angle

$$i_{pitch} = \arctan \left(\frac{F_{tidal}}{v_c^2/R} \right)$$

4. UQFF MUGE for M51

$$g_{M51} = \frac{G(M_a + M_b)}{r^2} (1 + H(z)) + \rho_{ISM} V g \frac{\rho_{SCm}}{\rho_{UA}}$$

5. UQFF Star Formation Efficiency

$$\text{SFE}_{UQFF} = \text{SFR} \left(1 + \frac{F_{tidal}}{g_{grav}} \right) (1 + f_{TRZ})$$

References

- Salo & Laurikainen (2000), MNRAS, 319, 393
- UQFF Framework, Star-Magic Session 174

Whitepaper auto-generated by _gen_whitepapers_688_701.py — Star-Magic Session 174